

CST34

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## DIGITAL SYSTEM DESIGN.

## Unit - I (Review of Binary number systems)

## Introduction (digital)

Computing and Electronics

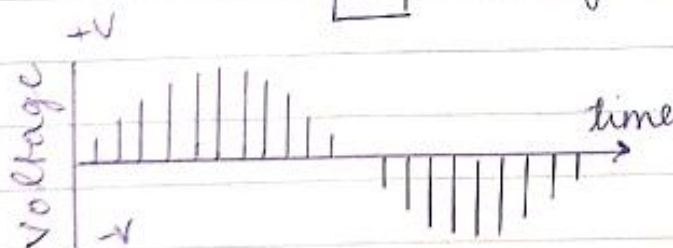
Real world information is converted to binary numeric form as in digital audio and digital photography.

## Difference between Digital and Analog

Digital - signals are represented either by 0 or 1.

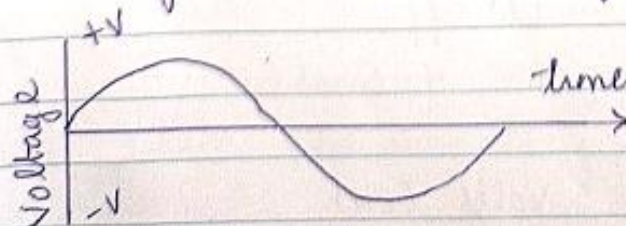
0  low signal

1  high signal



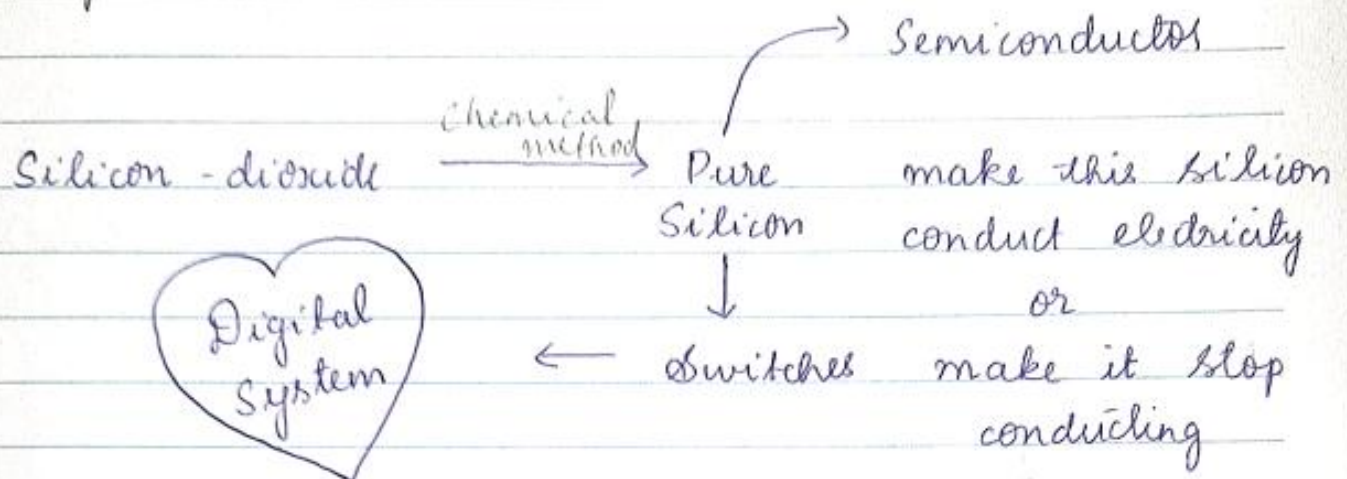
Digital representation.

Analog - signal vary continuously



analog representation

## Computers :-



1. Bunch of switches  $\rightarrow$  chip
  2. Bunch of chip  $\rightarrow$  circuit boards
  3. Bunch of circuit boards  $\rightarrow$  digital system
- Personal  $\leftarrow$  Computers

## Digital Electronics :-

1. Many digital systems are made by using these chips and circuit boards.
  2. VLSI - Very Large Scale Integration (chip designing)
  3. Computer systems are constructed of digital electronics
  4. Electronic circuits can only exist in one of two states. (a) On (b) Off
  5. Digital means zeros & Ones
- Example
- (a) Transistor with 5 Volts (ON)
  - (b) Transistor with no voltage (OFF)

another example CD-ROMs

Use microscopic dark spots on the surface of the disk (OFF)

Ordinary shiny surface (ON)

6. "On" and "Off" stored inside the computer are used to encode numbers using the binary number systems.

7. Computer's electronics can easily manipulate numbers stored in binary by treating 1 as "on" and 0 as "off".

### Digital System:-

1. Digital systems are nothing but electronic systems that use digital signals.

2. A digital system is an interconnection of digital modules and these digital modules are nothing but digital circuits.

3. Used in communications, business, medical treatment, internet and many other commercial, industrial and scientific enterprises

example:- Digital computers.

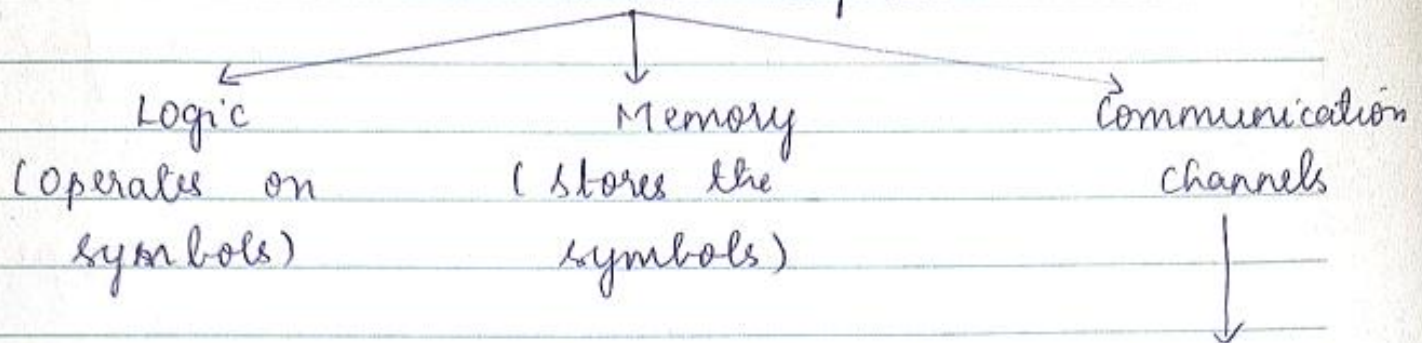
4. A digital system represents information in discrete symbols.

Two symbols  $\rightarrow$  binary digits.

$\left\{ \begin{array}{l} \rightarrow 0's \\ \rightarrow 1's \end{array} \right.$  All Information

5. Systems are constructed from three basic components.

## three basic components



(usually wires where information will move to the required place, so that values generated by logic, in one location, can be stored in memory)

6. Digital systems are being used more than analog systems.

Why we are using digital systems more when compared to analog systems?

1. Digital systems are accurate & reliable
2. Surprisingly fast and cheap as compared to analog systems

## Advantages

1. Easy to design
2. Store information easily
3. Less affected by noise

Number system:-

Decimal :- base 10 (has 10 digits)

0, 1, 2, ... 9

Binary :- base 2 (has 2 digital)

0 and 1.

Octal :- base 8 (has 8 digit)

0, 1, 2, ... 7

Hexadecimal :- base 16 (has 16 digit)

0, 1, 2, ... 9 & A, B, C ... F

| Decimal | Binary |   |   |   | Octal | Hexadecimal |
|---------|--------|---|---|---|-------|-------------|
| 0       | 0      | 0 | 0 | 0 | 0     | 0           |
| 1       | 0      | 0 | 0 | 1 | 1     | 1           |
| 2       | 0      | 0 | 1 | 0 | 2     | 2           |
| 3       | 0      | 0 | 1 | 1 | 3     | 3           |
| 4       | 0      | 1 | 0 | 0 | 4     | 4           |
| 5       | 0      | 1 | 0 | 1 | 5     | 5           |
| 6       | 0      | 1 | 1 | 0 | 6     | 6           |
| 7       | 0      | 1 | 1 | 1 | 7     | 7           |
| 8       | 1      | 0 | 0 | 0 | 10    | 8           |
| 9       | 1      | 0 | 0 | 1 | 11    | 9           |
| 10      | 1      | 0 | 1 | 0 | 12    | A           |
| 11      | 1      | 0 | 1 | 1 | 13    | B           |
| 12      | 1      | 1 | 0 | 0 | 14    | C           |
| 13      | 1      | 1 | 0 | 1 | 15    | D           |
| 14      | 1      | 1 | 1 | 0 | 16    | E           |
| 15      | 1      | 1 | 1 | 1 | 17    | F           |

## Decimal Number system?

10 - decimal base

example  $2586_{10} = (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0)$



0, 1, 2, 3 are positions of the digits

$$= 2000 + 500 + 80 + 6$$

## Binary to decimal Number system

example:-

$$\begin{aligned} 10101_2 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 16 + 0 + 4 + 0 + 1 \\ &= 21_{10} \end{aligned}$$

Try this:-

$$\begin{aligned} 1) \quad 1100_2 &= (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (0 \times 2^0) \\ &= 8 + 4 + 0 + 0 \\ &= 12_{10} \end{aligned}$$

$$\begin{aligned} 2) \quad 10011_2 &= (1 \times 2^4) + (0 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \\ &= 16 + 0 + 0 + 2 + 1 \\ &= 19_{10} \end{aligned}$$

## Octal to decimal Number system

example:-

$$\begin{aligned} 2057_8 &= (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0) \\ &= 1024 + 0 + 40 + 7 \\ &= 1071_{10} \end{aligned}$$

Try this:-

$$2057_8 = (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0)$$

$$\begin{aligned} 1) \quad 54318_8 &= (5 \times 8^4) + (4 \times 8^3) + (3 \times 8^2) + (1 \times 8^1) + (8 \times 8^0) \\ &= 20480 + 2048 + 192 + 8 + 8 \\ &= 22736 \end{aligned}$$

$$2) \overset{3}{2} \overset{2}{3} \overset{1}{5} \overset{0}{8}_8 = (2 \times 8^3) + (3 \times 8^2) + (5 \times 8^1) + (8 \times 8^0) \\ = 1024 + 192 + 40 + 8$$

Hexadecimal to Decimal Number system

example:  $\overset{2}{1} \overset{1}{A} \overset{0}{F}_{16} = (1 \times 16^2) + (A \times 16^1) + (F \times 16^0) \\ = 1 \times 256 + 10 \times 16 + 15 \times 1 \\ = 256 + 160 + 15 \\ = 431_{10}$

Try this:

$$\overset{4}{A} \overset{3}{B} \overset{2}{C} \overset{1}{D} \overset{0}{E}_{16} = (10 \times 16^4) + (11 \times 16^3) + (12 \times 16^2) + (13 \times 16^1) + (14 \times 16^0) \\ = 10 \times 65536 + 11 \times 4096 + 12 \times 256 + 13 \times 16 + 14 \times 1 \\ = 655360 + 45056 + 3072 + 208 + 14 \\ = 703,510$$

$$\overset{2}{3} \overset{1}{B} \overset{0}{4} \overset{-1}{.} \overset{-2}{C} \overset{-3}{8}_{16} = (3 \times 16^2) + (11 \times 16^1) + (4 \times 16^0) + (12 \times 16^{-1}) + (8 \times 16^{-2}) \\ = 3 \times 256 + 176 + 4 + 12 \times 0.0625 + 8 \times 0.00390625 \\ = 768 + 176 + 4 + 0.75 + 0.03125 \\ = 948.78125_{10}$$

Converting a Decimal Number to a Number of Another Base.

1) Decimal Number to Binary Number

example:  $12_{10} = ?_2$

|   |    |   |                 |
|---|----|---|-----------------|
| 2 | 12 | 0 | ↑<br>remainders |
| 2 | 6  | 0 |                 |
| 2 | 3  | 1 |                 |
|   |    | 1 |                 |

Hence,  $12_{10} = 1100_2$

## 2) Decimal Number to Octal Number

example:  $952_{10} = ?_8$

Hence,  $952_{10} = 1670_8$

$$\begin{array}{r} 8 \overline{) 9520} \\ \underline{8 \phantom{00} 119} \phantom{0} 7 \\ 8 \phantom{00} \underline{14} \phantom{0} 6 \\ \phantom{00} 1 \phantom{00} 1 \end{array}$$

## 3) Decimal Number to Hexadecimal Number

example  $952_{10} = ?_{16}$

$$59 \times 16 = 944$$

$$952 - 944 = 8$$

Hence  $952_{10} = 3B8_{16}$

$$\begin{array}{r} 16 \overline{) 952} \phantom{0} 8 \\ 16 \phantom{00} \underline{59} \phantom{0} B \\ \phantom{00} 3 \phantom{00} 3 \end{array}$$

$$3 \times 16 = 48 \quad 59 - 48 = 11(B)$$

## Binary Number to Octal Number

example:  $562_8 = ?_2$

8421

Convert each Octal digit to 3 bit binary digits

$$5_8 = 101_2 \quad 6_8 = 110_2 \quad 2_8 = 010_2$$

Combine the binary groups.

Hence,  $562_8 = 101 \ 110 \ 010_2$

$$\begin{array}{c} \text{Octal Number?} \quad \overset{2^4}{1} \overset{2^3}{0} \overset{2^2}{1} \mid \overset{2^4}{1} \overset{2^3}{1} \overset{2^2}{0} \mid \overset{2^4}{0} \overset{2^3}{1} \overset{2^2}{0} \\ \phantom{\text{Octal Number?}} \quad (5 \quad 6 \quad 2)_8 \end{array}$$

## Hexadecimal to binary Number

example:  $2AB_{16} = ?_2$

Convert each hexadecimal digit to a 4 bit binary number.

$$2_{16} = 0010_2$$

$$A_{16} = 1010_2$$

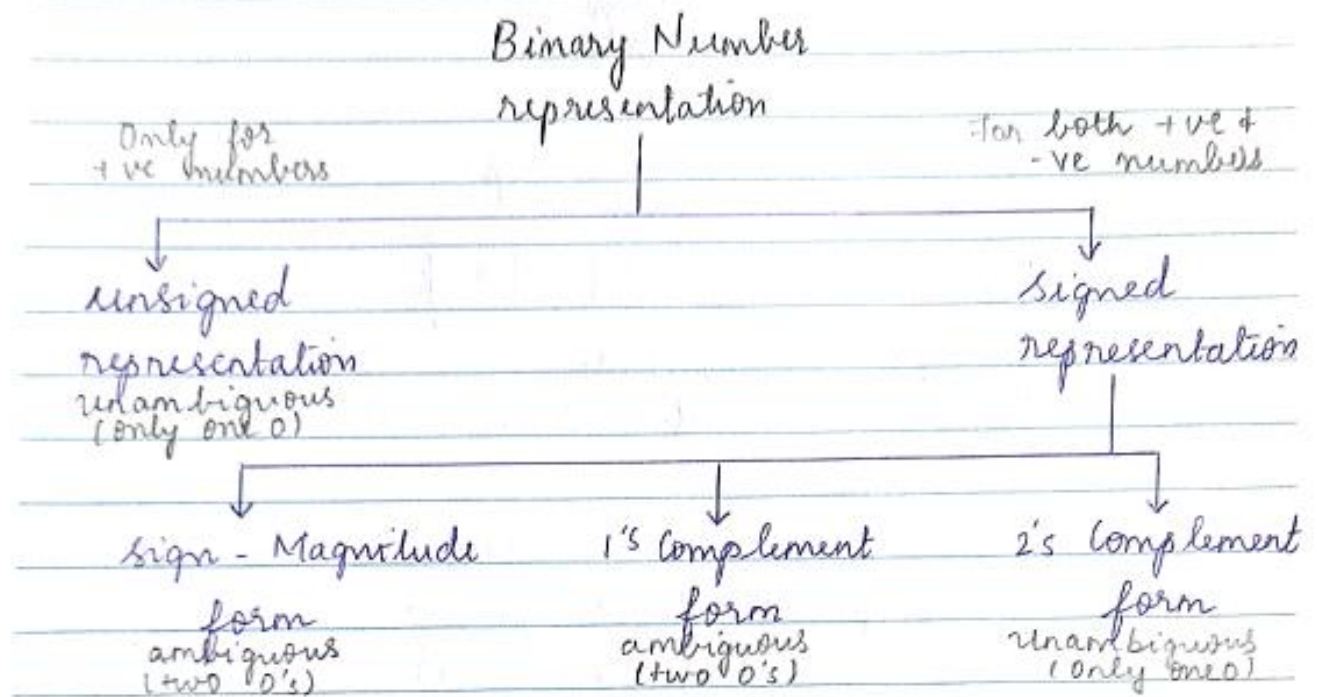
$$B_{16} = 1011_2$$

Combine the binary groups.

Hence,  $2AB_{16} = 0010 \ 1010 \ 1011$

Hexadecimal?  $\begin{matrix} 8 & 4 & 2 & 1 \\ 0 & 0 & 1 & 0 \end{matrix}$   $\begin{matrix} 8 & 4 & 2 & 1 \\ 1 & 0 & 1 & 0 \end{matrix}$   $\begin{matrix} 8 & 4 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{matrix}_2$   
 $(2 \quad A \quad B)_{16}$

## Signed Binary Numbers:-



## Unsigned Representation:-

In mathematics, positive numbers (including zero) are represented as unsigned numbers.

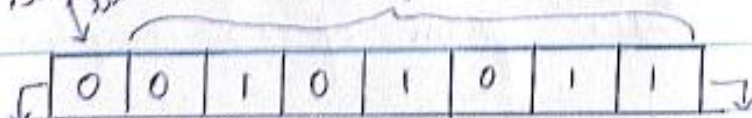
## Signed representation

Negative numbers we do use a -ve sign in front of the number.

### (i) Sign Magnitude

1 byte = 8 bit

Sign bit      Magnitude (Value)



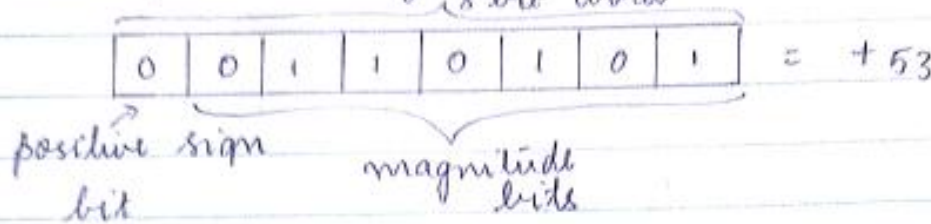
Most significant bit (MSB)

Least significant Bit (LSB)

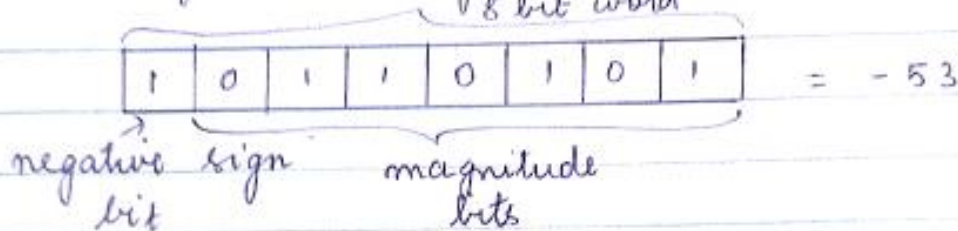
(0 represent that the number is positive)

"1's" and "0's", called a bit BINARY Digit

Positive signed binary Numbers



Negative signed binary Numbers.



$n-1$  bit signed binary number giving a reduced range of digits form:  
 $-2^{(n-1)}$  to  $+2^{(n-1)}$

4 bits to represent a signed binary number.

1-bit  $\rightarrow$  Sign bit

3 bits  $\rightarrow$  Magnitude bits

$$-2^{(4-1)-1}$$

$$\text{to } +2^{(4-1)-1}$$

$$-2^{(3)-1}$$

$$\text{to } +2^{(3)-1}$$

$$-7$$

to

$$+7$$

(ii) 1's Complement

1) By inverting each bit of a number.

2) The negative numbers can be represented in the form of 1's complement

## (iii) 2's Complement

By inverting each bit of a number and adding plus 1 to its least significant bit.

Example:

0 1 1 0 1 1 0      Original Binary Value

1 0 0 1 0 0 0 1      1's Complement

1 0 0 1 0 0 0 1

+ 1

1 0 0 1 0 0 1 0      2's Complement

## Arithmetic Operations

Binary arithmetic is a mathematical operation that involves performing calculations using the binary number system, which uses only two digits 0 and 1. Common binary arithmetic operations include addition, subtraction, multiplication and division.

### 1) Binary Addition

$$0 + 0 = 0$$

$$1 + 1 = (10)_2$$

$$0 + 1 = 1$$

$$1 + 1 + 1 = (111)_2$$

$$1 + 0 = 1$$

$$1 + 0 = (11)_2$$

$$1 + 1 = \overset{2}{1}0 \text{ (carry over the 1)}$$

|          |                                                                                                                                              |                                                   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Example: | $  \begin{array}{r}  1 \quad 0 \quad 1 \quad 0 \\  + 1 \quad 1 \quad 1 \quad 0 \\  \hline  1 \quad 1 \quad 0 \quad 0 \quad 0  \end{array}  $ | decimal $(10)_{10}$<br>$(14)_{10}$<br>$(24)_{10}$ |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|

|       |   |   |   |   |   |   |   | M  | T      | W | T     | F | S | S |
|-------|---|---|---|---|---|---|---|----|--------|---|-------|---|---|---|
| carry | 1 | 0 | 0 | 1 | 0 | 1 | 1 |    |        |   |       |   |   |   |
|       | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1) | 1101   | + | 1011  |   |   |   |
|       | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 2) | 10101  | + | 1111  |   |   |   |
|       |   |   |   |   |   |   |   | 3) | 100110 | + | 11011 |   |   |   |

## 2) Binary Subtraction:

$$1 - 0 = 1$$
$$1 - 1 = 0$$

Borrowing is required when subtracting 1 from 0, similar works in decimal subtraction

example:  $0 - 0 = 0$   $0 - 1 = 1$  with 1 borrow

1) Borrow 8 4 2 1

$$\begin{array}{r} 1 \quad 1 \quad 1 \quad 0 \quad \text{decimal} \quad (14)_{10} \\ - 0 \quad 0 \quad 1 \quad 1 \quad (-) (3)_{10} \\ \hline \end{array}$$

1 0 1 1 1 6 3 4 2 (11)<sub>10</sub>

$$\begin{array}{r} 2) \quad \begin{array}{cccc} 8 & 4 & 2 & 1 \\ 0 & 2 & 2 & 2 \\ \hline 0 & 0 & 0 & 0 \end{array} \\ \begin{array}{cccc} - & 0 & 1 & 1 \\ \hline 0 & 1 & 0 & 1 \end{array} \end{array}$$
$$\begin{array}{r} 3) \quad \overset{298}{0} \overset{294}{0} \overset{292}{0} \overset{291}{0} \overset{290}{0} \\ - \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \\ \hline 0 \quad 0 \quad 0 \quad 0 \quad 1 \end{array}$$

4)  $110101 - 10110$

5) 111010 - 1101

### 3) Binary Multiplication:-

Similar to decimal ~~division~~ multiplication but only using 0 and 1. Multiplying any number by 0 result in 0, and multiplying any number by 1 results in the original number.

### decimal Multiplication:-

25 (Multiplicand)

\* 11 (Multiplier)

25

25

275

$$\mathcal{O}_X \otimes \mathcal{O} = \mathcal{O}$$
$$0 \times 1 = 0$$
$$1 \times 0 = 0$$
$$1 \times 1 = 1$$

example

$$\begin{array}{r}
 \phantom{00}110 \\
 \times \phantom{00}101 \\
 \hline
 \phantom{00}110 \\
 \phantom{00}000 \\
 (+) \phantom{00}110 \\
 \hline
 11110
 \end{array}$$

decimal 6

$\times 5$

Partial product 30

Partial product

Partial product

$$16 + 8 + 4 + 2 = 30$$

2)

$$\begin{array}{r}
 \phantom{0000}1100 \\
 (\times) \phantom{0000}1101 \\
 \hline
 \phantom{0000}1100 \\
 \phantom{0000}0000x \\
 \phantom{0000}1100xx \\
 (+) \phantom{0000}1100xx \\
 \hline
 10011100
 \end{array}$$

128 64 32 16 8 4 2 1

$12 \times 13$

156

1)  $1101 \times 101$

2)  $111 \times 110$

3)  $1011 \times 1001$

$$128 + 16 + 08 + 04 = 156$$

Multiplication of fractional Binary Numbers.

1)  $110.01 - 2$

0.25

$11.10 - 1$

$\times 3.5$

$11001$

21.875

$11001x$

$11001xx$

$10101.111$

16 8 4 2 1 0.5 0.25 0.125

$$16 + 4 + 1 + 0.5 + 0.25 + 0.125$$

$$= 21.875$$

2)  $0111.1 - 1$

$7.5 \times 10.5$

$1010.1 - 1/2$

78.75

$01111$

$00000x$

$01111xx$

$00000xxx$

$11111xxx$

$1001110.11$



Binary division with fractional Numbers.

3)  $53.75 = 110101.110$

$02 = 10$

$11010.111 \rightarrow 26.875$

$10 \overline{) 110101.110} \rightarrow 53.75$

$\begin{array}{r}
 10 \overline{) 110101.110} \\
 \underline{10} \phantom{00000} \\
 010 \phantom{00000} \\
 \underline{10} \phantom{00000} \\
 0010 \phantom{0000} \\
 \underline{10} \phantom{0000} \\
 0011 \phantom{000} \\
 \underline{10} \phantom{000} \\
 011 \phantom{00} \\
 \underline{10} \phantom{00} \\
 010 \phantom{0} \\
 \underline{10} \phantom{0} \\
 00
 \end{array}$

## Binary Codes

Code is group of symbols, which is used to represent data.

It is a way of representing letters, characters and numbers using group of '1' and '0'.

Advantages:

- 1) Binary codes are suitable for the Computer applications, digital communications.
- 2) Binary codes make the analysis and designing of digital circuits.

# Classification of binary Codes.

## Numeric codes

Numeric codes are the codes which represent only numbers of 0's and 1's. The numeric codes which are used to represent the decimal digits are known as binary coded decimal (BCD) Gray Code, Excess-3

example: 8421, 5421, 2421 are BCD

## Alphanumeric codes

Alphanumeric codes are the codes which presents combination of both letters (alphabet) and decimal number as a sequence of 0's and 1's.

example: ASCII code

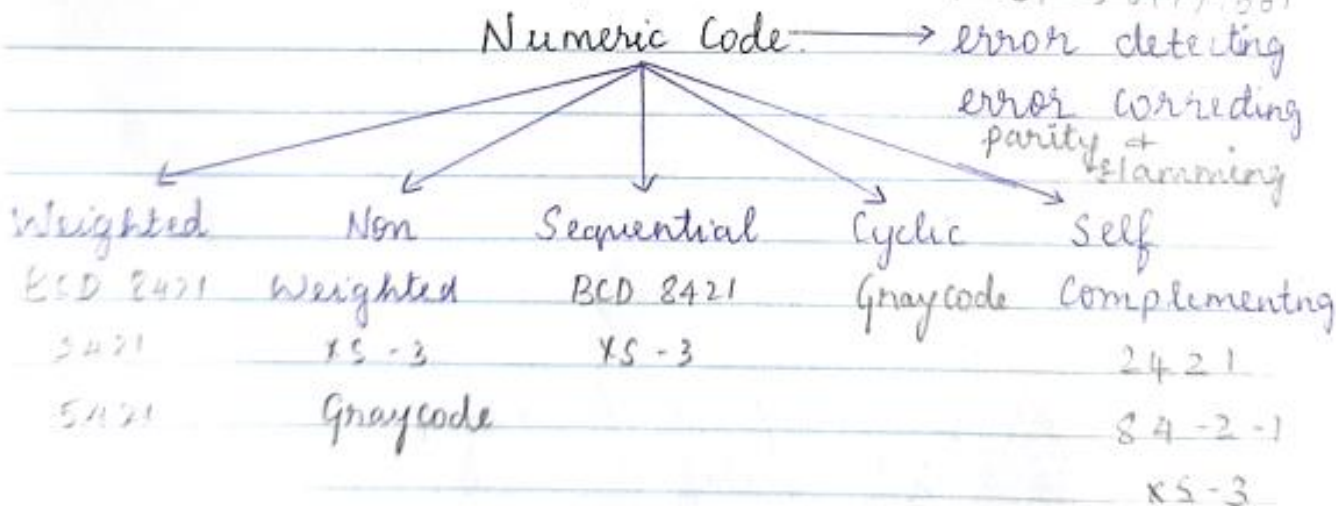
## EBCDIC code.

Extended Binary Coded Decimal Interchange Code.

It is basically 7 bit code. 27 = 128 symbols/code.

0 - 30 → 01110000  
1 - 39 → 01111001

## Numeric Code.



## Weighted binary Code:

Weighted codes are those codes which has positional value and position of each digit carry a certain or specific weight.

example: 8421, 2421

ii) each digit is represented by fixed number of bits

BCD code :: (i) type of binary code

BCD - Binary Coded decimal

iii) Normally 4 bits are used.

| Decimal | 8 | 4 | 2 | 1 | Weightage   |
|---------|---|---|---|---|-------------|
| 3       | 0 | 0 | 1 | 1 | $2 + 1 = 3$ |

Example :  $(23)_{10}$       2 - 0010      3 - 0011

$$(23)_{10} = 00100011$$

for fractional numbers.

$$(56.79)_{10} = 01010110.01111001$$

5      6      .      7      9

0101    0110    0111    1001

Hierarchical  $\Rightarrow (xx.xx)_b$  - decimal - BCD

$$\text{example: } (5C)_{16} \rightarrow 5 \times 16^1 + C \times 16^0$$

$$\rightarrow (92)_{10} = 10010010$$

9      2

1001    0010

Octal

$$\text{example } (621)_8 \rightarrow 6 \times 8^2 + 2 \times 8^1 + 1 \times 8^0$$

$$\rightarrow (401)_{10} = 0100 \ 0000 \ 0001$$

4      0      1

0100    0000    0001

Binary to BCD conversion

$$\text{example: } (11001010)_2$$

$$= 1 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= (202)_{10} = 0010 \ 0000 \ 0010$$

2      0      2

0010    0000    0010

Non-Weighted code:

Non-Weighted codes has no specific weight i.e., each digit position of a given number has no specific value.

Excess-3 codes & Gray code.

Self Complementing Code:

The codes representing the pair of complementary digits are also complementary to each other. 84-2-1, 2421

example: excess-3 code

241: 0101 0111 0100  $(9F)_{16} = 2421$

758: 1010 1000 1011  $(101101)_2 = 84-2-1$

Excess-3 code.

(7)

8421 BCD +3

Succeeding code

preceding code

Decimal  $\rightarrow$  8421 BCD  $\xrightarrow{+0011}$  XS+3 Code

example:

 $(23)_{10}$ 

2

3

0010

0011

+ 0011 + 0011

XS-3

0101

0110

 $(23)_{10} = 01010110$ 

$(xx.xx)_b \rightarrow$  decimal  $\rightarrow$  BCD 8421  
 $\xrightarrow{+0011}$  XS-3 code.

- (i) extended version of BCD code
- (ii) add +3 to BCD code.

|   | XS-3 |
|---|------|
| 0 | 0011 |
| 1 | 0100 |
| 2 | 0101 |
| 3 | 0110 |
| 4 | 0111 |
| 5 | 1000 |
| 6 | 1001 |
| 7 | 1010 |
| 8 | 1011 |
| 9 | 1100 |

hexadecimal:

$$(5C)_{16} \rightarrow 5 \times 16^1 + 12 \times 16^0$$

$$(92)_{10} \rightarrow 10010010$$

$$+ 0011 + 0011$$

$$XS-3 \quad 1100 \quad 0101$$

$$(5C)_{16} = 11000101$$

Octal.

$$(734)_8 = 7 \times 8^2 + 3 \times 8^1 + 4 \times 8^0$$

$$= 448 + 24 + 4 = (476)_{10}$$

$$4 \quad 7 \quad 6$$

$$0100 \quad 0111 \quad 0110$$

$$+ 0011 + 0011 + 0011$$

$$XS-3 \quad 0111 \quad 1010 \quad 1001$$

$$(734)_8 = 011110101001$$

Sequential Code:

In sequential codes each succeeding code is one binary number greater than its preceding code.

XS-3 code  $\rightarrow$  add +0001 to get next value.

Gray code (i) error correcting codes

(ii) should be change of 1 bit from 1 step to another

|   |      |    |      |     |      |
|---|------|----|------|-----|------|
| 0 | 0000 | 7  | 0100 | 14  | 1001 |
| 1 | 0001 | 8  | 1100 | 15  | 1000 |
| 2 | 0011 | 9  | 1101 | xor |      |
| 3 | 0010 | 10 | 1111 | 0   | 0 0  |
| 4 | 0010 | 11 | 1110 | 0   | 1 1  |
| 5 | 0111 | 12 | 1010 | 1   | 0 1  |
| 6 | 0101 | 13 | 1011 | 1   | 1 0  |

Binary to Gray Conversion

| Binary |       |       |       | Gray  |       |       |       |
|--------|-------|-------|-------|-------|-------|-------|-------|
| $B_3$  | $B_2$ | $B_1$ | $B_0$ | $G_3$ | $G_2$ | $G_1$ | $G_0$ |
| 1      | 0     | 1     | 0     | 1     | 1     | 1     | 1     |

Note :-  $G_3 = B_3$

$$G_2 = B_3 \otimes B_2 \quad 1 \otimes 0 = 1$$

$$G_1 = B_2 \otimes B_1 \quad 0 \otimes 1 = 1$$

$$G_0 = B_1 \otimes B_0 \quad 1 \otimes 0 = 1$$

| Binary |       |       |       | Gray  |       |       |       |
|--------|-------|-------|-------|-------|-------|-------|-------|
| $B_3$  | $B_2$ | $B_1$ | $B_0$ | $G_3$ | $G_2$ | $G_1$ | $G_0$ |
| 0      | 1     | 0     | 0     | 0     | 1     | 1     | 0     |

Gray to Binary Conversion

| Gray  |       |       |       | Binary |       |       |       |
|-------|-------|-------|-------|--------|-------|-------|-------|
| $G_3$ | $G_2$ | $G_1$ | $G_0$ | $B_3$  | $B_2$ | $B_1$ | $B_0$ |
| 1     | 1     | 1     | 0     | 1      | 0     | 1     | 1     |

Note :-  $B_3 = G_3 = 1$

$$B_2 = B_3 \otimes G_2$$

$$B_1 = B_2 \otimes G_1$$

$$B_0 = B_1 \otimes G_0$$

| Gray  |       |       |       | Binary |       |       |       |
|-------|-------|-------|-------|--------|-------|-------|-------|
| $G_3$ | $G_2$ | $G_1$ | $G_0$ | $B_3$  | $B_2$ | $B_1$ | $B_0$ |
| 0     | 1     | 0     | 1     | 0      | 1     | 1     | 0     |

Application

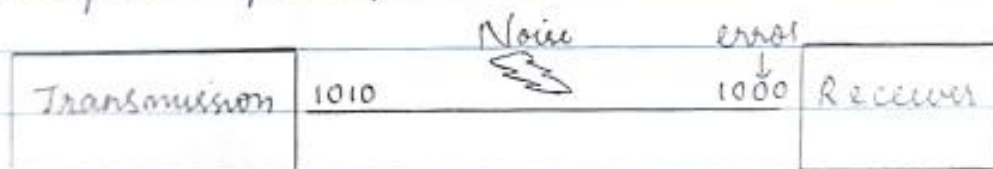
(i) Instrumentation (ii) Communication systems

- 1) Gray code is popularly used in the shaft position encoders.
- 2) A shaft position encoder produces a code word which represents the angular position of the shaft.

### Error detection code

Whenever a message is transmitted, it may get scrambled by noise or data may get corrupted. To avoid this, we use error-detecting codes which are additional data added to a given digital message to help us detect if an error occurred during transmission of the message.

Example: parity check. (CRC, checksum)



parity a parity bit is an extra bit which is added with a message to make parity bit.

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | x |
|---|---|---|---|---|---|---|---|

the total no. of one's either even / odd.

Odd parity: Total no. of 1's in the code,

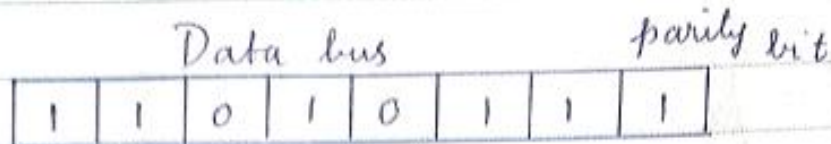
including the parity bit should be odd

|          |   |   |   |   |   |   |            |
|----------|---|---|---|---|---|---|------------|
| Data bit |   |   |   |   |   |   | parity bit |
| 1        | 0 | 1 | 1 | 0 | 0 | 1 | 1          |

Even parity Total no. of 1's in the code.

including the parity bit should be even

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|



Let's consider Odd parity



Let's consider even parity



Advantage:

Simplicity  
easy to use

Disadvantage:

It may fail to catch errors  
if two data bits are corrupted, for instance, parity will not detect the errors.

Error Correction Code

Error correction codes are generated by using the specific algorithms used for removing and detecting errors from the message transmitted over noisy channels.

The error correcting codes find the correct number of bits and their positions in the message.

The hamming Code technique is used for error correction.

(i) Given by R.W Hamming

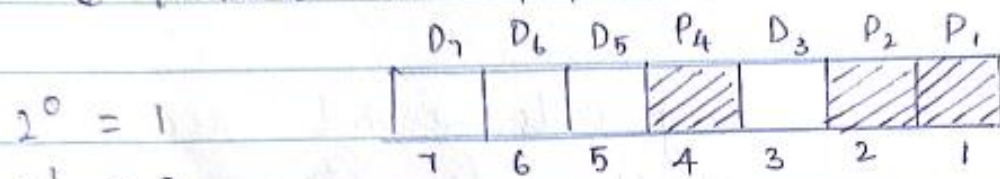
(ii) Easy to implement

(iii) 7 bit hamming code is used commonly

Data bits - 4

Parity bits - 3

$2^n$  {where  $n = 0, 1, \dots, n\}$



$$2^2 = 4$$

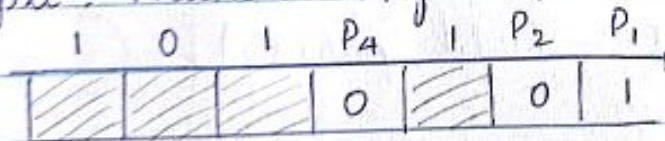
$$2^3 = 8 \text{ (7 bit no } 8^{\text{th}} \text{ position)}$$

$$P_1 = D_3 D_5 D_7$$

$$P_2 = D_3 D_6 D_7$$

$$P_4 = D_5 D_6 D_7$$

Example: Transmitting 4 bit to a receiver even parity



To determine the values  $P_1, P_2, P_4$

$$P_1 = 1 \ 1 \ 1 \Rightarrow P_1 = 1$$

$$P_2 = 1 \ 0 \ 1 \Rightarrow P_2 = 0$$

$$P_4 = 1 \ 0 \ 1 \Rightarrow P_4 = 0$$

$\Rightarrow$  Receiver receives 1010101

how it will detect the error.

so receiver receives 1110101 instead  
1010101

$$P_1 = 1 \quad \begin{matrix} D_3 & D_5 & D_7 \\ \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \end{matrix}$$

$$P_2 = 0 \quad \begin{matrix} D_3 & D_6 & D_7 \\ \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \end{matrix} \quad (\text{There is an error})$$

$$P_4 = 0 \quad \begin{matrix} D_5 & D_6 & D_7 \\ \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \end{matrix}$$

How to correct the error.

If the 7 bit hamming code word received by a receiver is 1011011. Assuming the even parity state whether the received code word is correct or wrong. If wrong locate the bit having error.

Solution:-

| $D_7$ | $D_6$ | $D_5$ | $P_4$ | $D_3$ | $P_2$ | $P_1$ |
|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0     | 1     | 1     | 0     | 1     | 1     |

↓  
error

| $P_4$ | $D_5$ | $D_6$ | $D_7$ |                        |
|-------|-------|-------|-------|------------------------|
| 1     | 1     | 0     | 1     | (odd parity) $P_4 = 1$ |

| $P_2$ | $D_3$ | $D_6$ | $D_7$ |                  |
|-------|-------|-------|-------|------------------|
| 1     | 0     | 0     | 1     | (even) $P_2 = 0$ |

| $P_1$ | $D_3$ | $D_5$ | $D_7$ |                        |
|-------|-------|-------|-------|------------------------|
| 1     | 0     | 1     | 1     | (odd parity) $P_1 = 1$ |

$$P_4 \ P_2 \ P_1 = (101)_2 = (5)_{10}$$

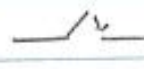
having error

The correct code

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|

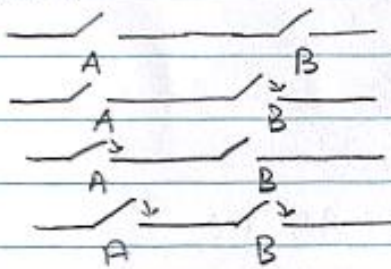
## Boolean Algebra.

1. Boolean Variable: Two value, true/false  
or 1/0

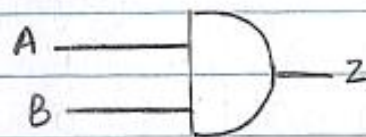
2. Basic Operations  closed 1 high V  
open 0 low voltage

- AND
- OR
- NOT

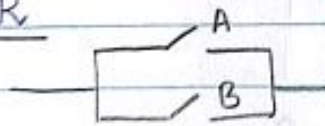
### AND



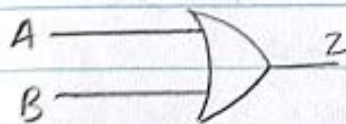
| A | B | $Z = A \cdot B$ / $Z = AB$ |
|---|---|----------------------------|
| 0 | 0 | 0                          |
| 0 | 1 | 0                          |
| 1 | 0 | 0                          |
| 1 | 1 | 1                          |



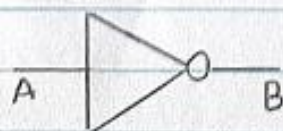
### OR



| A | B | $Z = A + B$ |
|---|---|-------------|
| 0 | 0 | 0           |
| 1 | 0 | 1           |
| 0 | 1 | 1           |
| 1 | 1 | 1           |



### NOT



| A | $B = A' = \bar{A}$ |
|---|--------------------|
| 0 | 1                  |
| 1 | 0                  |

### 3. Boolean function

-  $F = f(a, b, c, \dots)$

- Boolean expression - boolean variables + operations

### 4. Truth table

- Value of boolean function for every possible combination of input

- Two expressions are same, if same value for all combinations.

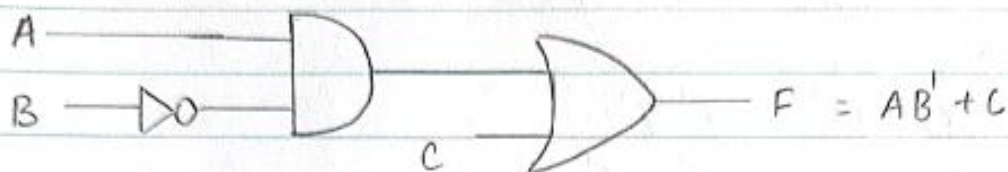
Boolean function example:

$$F = AB' + C = (A + C)(B' + C)$$

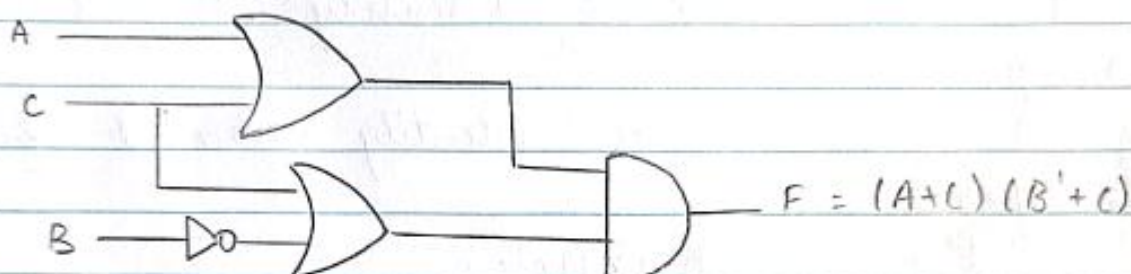
| A | B              | C | $AB'$ | $F = AB' + C$ |
|---|----------------|---|-------|---------------|
| 0 | 0'             | 0 | 0     | 0             |
| 0 | 0'             | 1 | 0     | 1             |
| 0 | 1 <sup>0</sup> | 0 | 0     | 0             |
| 0 | 1 <sup>0</sup> | 1 | 0     | 1             |
| 1 | 0'             | 0 | 1     | 1             |
| 1 | 0'             | 1 | 1     | 1             |
| 1 | 1 <sup>0</sup> | 0 | 0     | 0             |
| 1 | 1 <sup>0</sup> | 1 | 0     | 1             |

| $(A + C)$ | $(B' + C)$ | $(A + C)(B' + C) = F$ |
|-----------|------------|-----------------------|
| 0         | 1          | 0                     |
| 1         | 1          | 1                     |
| 0         | 0          | 0                     |
| 1         | 1          | 1                     |
| 1         | 1          | 1                     |
| 1         | 1          | 1                     |
| 1         | 0          | 0                     |
| 1         | 1          | 1                     |

$$F = AB' + C$$



$$F = (A+C)(B'+C)$$



### Basic Theorems:-

#### 1. Operations with 0 and 1

$$X + 0 = X$$

$$X + 1 = 1$$

$$X \cdot 1 = X$$

$$X \cdot 0 = 0$$

#### 2. Idempotent Law

$$X + X = X$$

$$X \cdot X = X$$

#### 3. Involution Law

$$(X')' = X$$

#### 4. Law of Complementarity.

$$X + X' = 1$$

$$X \cdot X' = 0$$

#### 5. Commutative

$$XY = YX$$

$$X + Y = Y + X$$

#### 6. Associative

$$(XY)Z = X(YZ) = XYZ$$

$$(X+Y)+Z = X+(Y+Z) = X+Y+Z$$

Law of Boolean algebra

7. Distributive

$$x(y+z) = xy + xz$$

$$x + yz = (x+y)(x+z)$$

8. DeMorgan's Law

$$(x+y)' = x'y'$$

$$(xy)' = x' + y'$$

Duality principle

1) Element 0 is identity element for + operator (OR)

2) Element 1 is identity element for . operator (AND)

3) Duality principle:

- Law of boolean algebra remain same if operators and identity elements are interchanged.

4) DeMorgan's duality:-

- a boolean expression remains same if all variable, operators are replaced with its dual.

Simplification theorems.

1. Uniting

$$a) xy + xy' = x$$

using distributive property  
 $= x(y + y')$

$$= x$$

$$b) (x+y)(x+y') = x$$

using distributive property

$$= x + yy'$$

$$= x + 0$$

$$= x$$

## 2. Absorption

a)  $x + xy = x$

$$= x \cdot 1 + xy$$

$$= x(1+y)$$

$$= x \cdot 1$$

b)  $x(x+y) = x$

$$= xx + xy$$

$$= x + xy$$

$$= x \cdot 1$$

## 3. Elimination

a)  $x + x'y = x + y$

$$= (x+x')(x+y)$$

$$= 1 \cdot (x+y)$$

$$= (x+y)$$

b)  $x(x'+y) = xy$

$$= x \cdot x' + x'y$$

$$= 0 + xy$$

$$= xy$$

## 4. Consensus

a)  $xy + x'z + yz = xy + x'z$

$$xy + x'z + yz(x+x')$$

$$xy + xyx + x'z + x'yx$$

$$xy(1+x) + x'z(1+y)$$

$$xy + x'z$$

Example:- absorption

1)  $Z = A'BC + A'$

$$A'(1+BC)$$

2)  $Z = \underbrace{[A+B'C]}_X + \underbrace{[D+EF]}_Y \underbrace{[A+B'C+(D+EF)']}_X$

$$= (X+Y)(X+Y') = X$$

$$Y = A+B'C$$

## SOP and POS representation

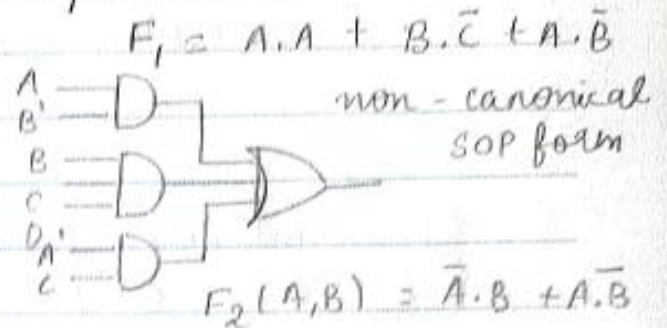
Two standard boolean expression

1) Sum of product

example

term  $AB' + BCD + A'C$

term  $A' + B' + C' + DE$

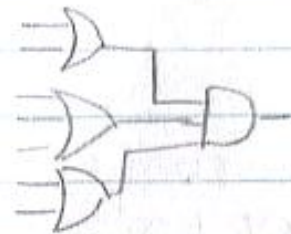


2) Products of sum:

example

$(A+B')(B+C+D)(A'+C)$

$A'B'C'(D+E)$



example

$(A+B) \underline{CD + DE}$  - neither SOP/POS

POS      SOP

Creating SOP : Multiplying out

1).  $(A+BC)(A+D+E)$

$$= A + AD + AE + BCA + BCD + BCE$$

$$= A(1 + D + E) + BCA + BCD + BCE$$

$$= A + BCA + BCD + BCE$$

$$= A + BCD + BCE$$

using  $(x+y)(x+z) = x + yz$

$$Y = A \quad Y = BC \quad Z = D + E$$

Sum of product (SOP)

1. Sum of product expression consists of two or more product terms (AND) that are OR'ed together.

2. Each product term consists of one or more literals appearing in either complemented or uncomplemented form.

3. The sum of product form is known as Disjunction Normal form.

example:  $F(A, B, C) = ABC + BCA + CAB$

4. The logical sum of those products that produce output "1" in the truth table

5. The corresponding logic circuit is an AND - OR circuit or NAND - NAND circuit.

Construct SOP Expression and logic gates from the given truth table.

| A | B | C | F |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

Note:- 1 - Uncomplement  
0 - complement

$\leftarrow \bar{A}\bar{B}C$

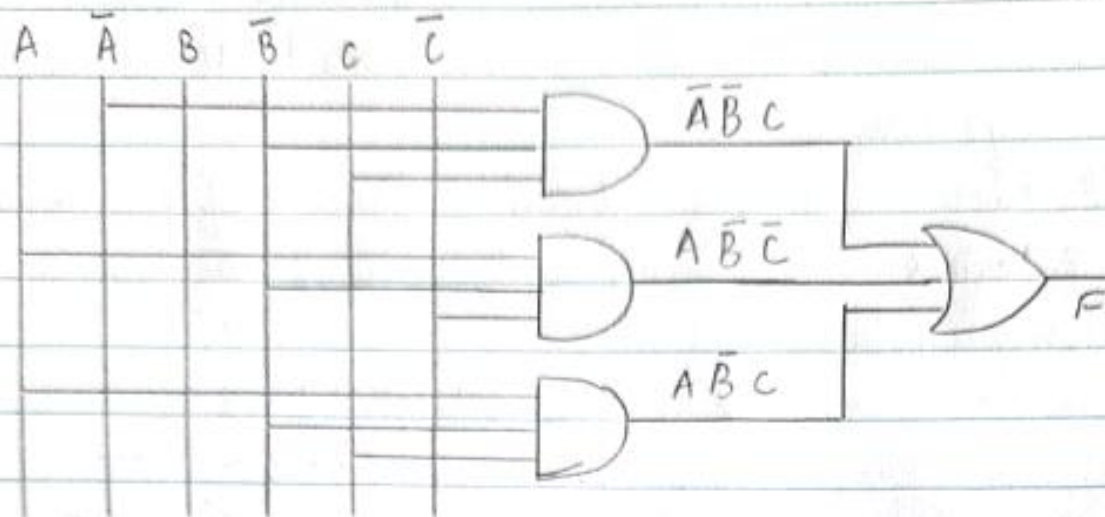
$\leftarrow A\bar{B}\bar{C}$

$\leftarrow A\bar{B}C$

The SOP Expression is

$$F = \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$$

## AND-OR circuit



2. Draw the logic gates for the given truth table using sum of products.

| A | B | C | Y |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

$\leftarrow \bar{A}\bar{B}\bar{C}$

$\leftarrow \bar{A}\bar{B}C$

$\leftarrow \bar{A}B\bar{C}$

$\leftarrow \bar{A}BC$

$\leftarrow ABC$

3. Suppose a 3 Variable truth table has a high output for these input conditions 000, 010, 100 and 110. What is the sum of product (SOP) form?

4. Draw a truth table and simplify the boolean expression for  $F(a, b, c) = \sum m(1, 3, 5, 7)$

$\Sigma$  - sum of product.

m - minterm - represents exactly one combination in the truth table

|   | a | b | c | f |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 0 | 1 | 1 | 1 |
| 4 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 1 |
| 6 | 1 | 1 | 0 | 0 |
| 7 | 1 | 1 | 1 | 1 |

denoted by  $m_j$ , where  $j$  is the decimal equivalent

of the minterm's corresponding binary

combination (bj)

0 - complemented

1 - uncomplemented

$$f = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$$

### Product of Sum (POS)

1. Product of sum expression consists of two or more sum terms that are AND<sup>ed</sup> together.

2. The product of sum form is known as Conjunctive normal form.

example:  $F(A, B, C) = (A+B+C)(\bar{B}+\bar{C}+A)$

3. The logical product of those sums that produce output 0's in the truth table

4. The logic gates is an OR-AND circuits or NOR-NOR circuits

1) Construct the expression is logic gates from the given truth table.

| A | B | C | Y |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

$$A + B + C$$

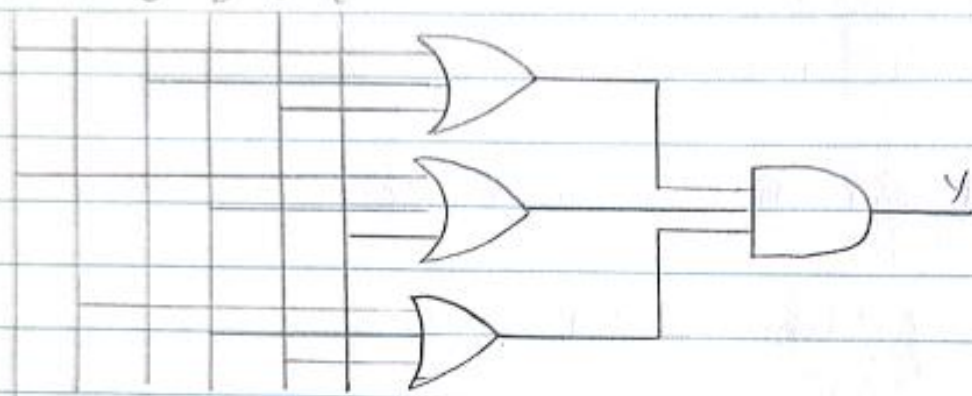
$$A + B' + C'$$

$$A' + B' + C$$

$$f = (A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$$

OR - AND circuit

A  $\bar{A}$  B  $\bar{B}$  C  $\bar{C}$



2) Suppose a truth table has a low output for the first three input conditions 000, 001, 010. If all other inputs are high. What is the product of sum (POS) form.

3) Draw a logic circuit for a function  $F(A, B, C) = \Pi M(0, 1, 2, 3)$

$\Pi$  - product of sum

M - Maxterms. represents exactly one combination in the truth table

Denoted by  $M_j$  where  $j$  is the decimal equivalent of the maxterms corresponding binary combination  $bj$

1 - complemented    0 - uncomplemented

$$f = (A+B+C)(A+B+\bar{C})(A+\bar{B}C)(A+\bar{B}+\bar{C})$$

## Canonical form

Every function  $F(A, B, C)$  has two canonical form

- 1) Canonical sum of Products (sum of minterms)
- 2) Canonical Product of Sums (sum of maxterms)

## Standard SOP or Minterm Canonical form

1) In SOP, if every product term involves every literals or its complement, then it becomes standard SOP form.

2) Each individual term in the standard SOP form is called **Minterms**

3) Therefore canonical SOP form is also known as **Minterm Canonical form**.

$$F(A, B, C) = \bar{A}\bar{B}C + A\bar{B}C + \bar{A}B\bar{C}$$

(Each product term consists of all literals in either complemented form or uncomplemented form.)

Rules to convert expressions in standard SOP form.

1) Find missing literals in each product term.

2) AND each term with missing literal and its complement.

$$F(A, B, C) = AC \boxed{CB + \bar{B}}$$

3) Expand the terms and reorder the literals.

4) Omit repeated product terms.

1) Convert the given expression in standard SOP form.

$$F(A, B, C) = AC + AB + BC$$

↳ A is missing  
→ C is missing  
→ B is missing

$$\begin{aligned} F(A, B, C) &= AC(B + \bar{B}) + AB(C + \bar{C}) + BC(A + \bar{A}) \\ &= ABC + A\bar{B}C + ABC + AB\bar{C} + ABC + \bar{A}BC \\ &= ABC + A\bar{B}C + AB\bar{C} + \bar{A}BC \end{aligned}$$

$$F(A, B, C) = ABC + A\bar{B}C + AB\bar{C} + \bar{A}BC$$

Sum of minterms =  $\sum m(3, 5, 6, 7)$

2) Convert the given expression in standard SOP form.

$$F(A, B, C) = A + ABC$$

↳ B + C is missing

$$\begin{aligned} F(A, B, C) &= A(B + \bar{B})(C + \bar{C}) + ABC \\ &= (AB + A\bar{B})(C + \bar{C}) + ABC \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} \end{aligned}$$

$$F(A, B, C) = \overset{7}{ABC} + \overset{6}{AB\bar{C}} + \overset{5}{A\bar{B}C} + \overset{4}{A\bar{B}\bar{C}}$$

$\sum m(4, 5, 6, 7)$

Standard POS form or maxterm Canonical form.

1) In product of sums, if every sum term involves every literal or its complement, then it becomes standard POS form.

2) Each individual term in the standard POS form is called **Maxterm**.

3) Therefore canonical POS form is also known as **Maxterm Canonical form**.

$$F(A, B, C) = (A + B + C)(A + \bar{B} + C)$$

(Each sum term consists of all literals in either complemented form or uncomplemented form)

Rules to convert expression in standard POS form

1) Finding Missing literals in each sum terms.

2) OR each sum term with missing literal and its complement.

3) Expand the literals and reorder literals.

4) Omit repeated sum terms.

1) Convert the given expression in standard POS form.

$$F(A, B, C) = (A+B)(B+C)(A+C)$$

└ B is missing

└ A is missing

└ C is missing

$$\begin{aligned} F &= [(A+B) + (C + \bar{C})] [(B+C) + (A \cdot \bar{A})] [(A+C) + (B \cdot \bar{B})] \\ &= (A+B+C) (A+B+\bar{C}) (A+B+C) (\bar{A}+B+C) \\ &\quad (A+B+C) (A+\bar{B}+C) \end{aligned}$$

$$= (A+B+C) (A+B+\bar{C}) (\bar{A}+B+C) (A+\bar{B}+C)$$

$$F(A, B, C) = (\overset{0}{A} + \overset{0}{B} + \overset{0}{C}) (\overset{0}{A} + \overset{0}{B} + \overset{1}{\bar{C}}) (\overset{1}{\bar{A}} + \overset{0}{B} + \overset{0}{C}) (\overset{1}{A} + \overset{1}{\bar{B}} + \overset{0}{C})$$

$$\text{product of maxterms} = \Pi M(0, 1, 2, 4)$$

2) Convert the given expression in standard POS form

$$Y(A, B, C) = A(A+B+C)$$

└ B & C Missing

$$Y(A, B, C) = [A + (B \cdot \bar{B}) + (C \cdot \bar{C})] (A+B+C)$$

$$= [(A+B) (A+\bar{B}) + (C \cdot \bar{C})] (A+B+C)$$

$$= [(A+B+C) (A+B+\bar{C}) (A+\bar{B}+C) (A+\bar{B}+\bar{C})] (A+B+C)$$

$$Y(A, B, C) = (A+B+C) (A+B+\bar{C}) (A+\bar{B}+C) (A+\bar{B}+\bar{C})$$

$$\text{product of maxterm} = \Pi M(0, 1, 4, 3)$$

## Karnaugh's Map

1) Used for minimizing boolean expression

2) Graphical representation of Truth table

3) It contains cell which holds minterm or max terms.

## Different types of K-Map

- 1) 2 - Variable k-map
- 2) 3 - Variable k-map
- 3) 4 - Variable k-map

### 2 Variable k-map.

2 Variables = A, B

2 Variables =  $2^n = 2^2$

2 Variables = 4 cells.

0 0 - 0

0 1 - 1

1 0 - 2

1 1 - 3

Note: Two cells are said to be adjacent if variables of minterms have only one variable changed

### Decimal Representation

| A \ B | 0 | 1 |
|-------|---|---|
| 0     | 0 | 1 |
| 1     | 2 | 3 |

### Binary representation

| A \ B | 0  | 1  |
|-------|----|----|
| 0     | 00 | 01 |
| 1     | 10 | 11 |

### Variable Representation

| A \ B     | $\bar{A}$        | A          |
|-----------|------------------|------------|
| $\bar{B}$ | $\bar{A}\bar{B}$ | $A\bar{B}$ |
| B         | $\bar{A}B$       | $AB$       |

0 - C

1 - UC

### 3 Variable k-map

3 Variables = A, B, C

3 Variables =  $2^n = 2^3$

3 Variables = 8 cells

000 - 0    110 - 6

001 - 1    111 - 7

010 - 2

011 - 3

100 - 4

101 - 5

## Decimal Representation

## Method-2

### Method - 1

| A \ BC | 00 | 01 | 10 | 11 |
|--------|----|----|----|----|
| 0      | 0  | 1  | 3  | 2  |
| 1      | 4  | 5  | 7  | 6  |

| AB \ C | 0 | 1 |
|--------|---|---|
| 00     | 0 | 1 |
| 01     | 2 | 3 |
| 11     | 6 | 7 |
| 10     | 4 | 5 |

## Binary Representation

| A \ BC | 00  | 01  | 11  | 10  |
|--------|-----|-----|-----|-----|
| 0      | 000 | 001 | 011 | 010 |
| 1      | 100 | 101 | 111 | 110 |

| AB \ C | 0   | 1   |
|--------|-----|-----|
| 00     | 000 | 001 |
| 01     | 010 | 011 |
| 11     | 110 | 111 |
| 10     | 100 | 101 |

## Variable representation

| A \ BC    | $\bar{B}\bar{C}$        | $\bar{B}C$        | $BC$              | $B\bar{C}$  |
|-----------|-------------------------|-------------------|-------------------|-------------|
| $\bar{A}$ | $\bar{A}\bar{B}\bar{C}$ | $\bar{A}\bar{B}C$ | $\bar{A}B\bar{C}$ | $\bar{A}BC$ |
| A         | $A\bar{B}\bar{C}$       | $A\bar{B}C$       | $AB\bar{C}$       | $ABC$       |

| AB \ C           | $\bar{C}$               | C                 |
|------------------|-------------------------|-------------------|
| $\bar{A}\bar{B}$ | $\bar{A}\bar{B}\bar{C}$ | $\bar{A}\bar{B}C$ |
| $\bar{A}B$       | $\bar{A}B\bar{C}$       | $\bar{A}BC$       |
| $A\bar{B}$       | $A\bar{B}\bar{C}$       | $A\bar{B}C$       |
| $AB$             | $AB\bar{C}$             | $ABC$             |

## 4 Variable K-map

4 Variable = A, B, C, D

4 Variable =  $2^n = 2^4$

4 Variable = 16 Cells

## Decimal Representation

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 0  | 1  | 3  | 2  |
| 01      | 4  | 5  | 7  | 6  |
| 11      | 12 | 13 | 15 | 14 |
| 10      | 8  | 9  | 11 | 10 |

## Binary Representation

| AB \ CD | 00   | 01   | 11   | 10   |
|---------|------|------|------|------|
| 00      | 0000 | 0001 | 0011 | 0010 |
| 01      | 0100 | 0101 | 0111 | 0110 |
| 11      | 1100 | 1101 | 1111 | 1110 |
| 10      | 1000 | 1001 | 1011 | 1010 |

## Variable representation

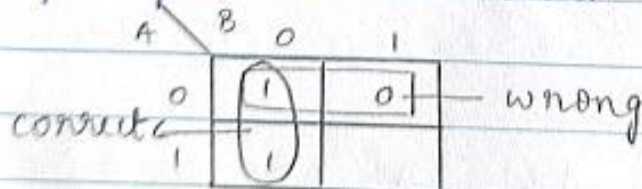
| AB \ CD          | $\bar{C}\bar{D}$               | $\bar{C}D$               | $C\bar{D}$               | $CD$               |
|------------------|--------------------------------|--------------------------|--------------------------|--------------------|
| $\bar{A}\bar{B}$ | $\bar{A}\bar{B}\bar{C}\bar{D}$ | $\bar{A}\bar{B}\bar{C}D$ | $\bar{A}\bar{B}C\bar{D}$ | $\bar{A}\bar{B}CD$ |
| $\bar{A}B$       | $\bar{A}B\bar{C}\bar{D}$       | $\bar{A}B\bar{C}D$       | $\bar{A}B C\bar{D}$      | $\bar{A}B CD$      |
| $AB$             | $AB\bar{C}\bar{D}$             | $AB\bar{C}D$             | $AB C\bar{D}$            | $AB CD$            |
| $A\bar{B}$       | $A\bar{B}\bar{C}\bar{D}$       | $A\bar{B}\bar{C}D$       | $A\bar{B} C\bar{D}$      | $A\bar{B} CD$      |

## Rules of Grouping / Filling of Cells SOP - 1

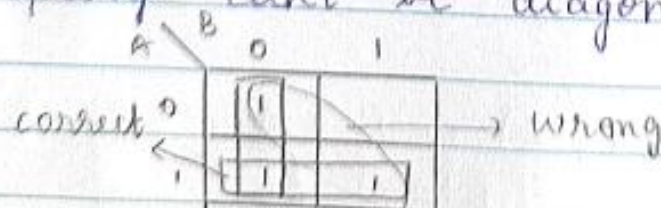
POS - 0

1) Adjacent cells which have 1's can be grouped together

2) Group can't contain "0"



3) Group can't be diagonal



note: overlapping is allowed.

4) Groups must contain  $2^n$  cells

5) Each group should be large as possible.

pairs - pair of 1's

Quads - ~~pair~~ group of four 1's

Octal - Group of eight 1's

1) Simplify the following expression using K-map.

$$F(A, B, C, D) = \sum_{(SOP)} m(4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

4 Variables  $(A, B, C, D) = 2^4 = 2^4 = 16$  cells.

| AB \ CD          | $\bar{C}\bar{D}$ | $\bar{C}D$      | $CD$            | $C\bar{D}$      |    |
|------------------|------------------|-----------------|-----------------|-----------------|----|
| $\bar{A}\bar{B}$ | 0 <sub>0</sub>   | 0 <sub>1</sub>  | 0 <sub>3</sub>  | 0 <sub>2</sub>  | 16 |
| $\bar{A}B$       | 1 <sub>4</sub>   | 1 <sub>5</sub>  | 1 <sub>7</sub>  | 1 <sub>6</sub>  | 8  |
| $AB$             | 1 <sub>12</sub>  | 1 <sub>13</sub> | 1 <sub>15</sub> | 1 <sub>14</sub> | 4  |
| $A\bar{B}$       | 1 <sub>8</sub>   | 1 <sub>9</sub>  | 1 <sub>11</sub> | 1 <sub>10</sub> | 2  |
|                  |                  |                 |                 |                 | 1  |

Group 1: 8  
Group 2: 4

Group 1:  $(AB) \quad \bar{A}B \cdot AB = B \cdot B = B$

$(CD) \quad \bar{C}\bar{D} \cdot \bar{C}D \cdot CD \cdot C\bar{D} = 1 \cdot 1 \cdot B = B$

Group 2:  $(AB) \quad A\bar{B} \cdot A\bar{B} = A \cdot A = A$

$(CD) \quad \bar{C}\bar{D} \cdot \bar{C}D \cdot CD \cdot C\bar{D} = 1 \cdot 1 \cdot A = A$

$$F(A, B, C, D) = B + A$$

$$2) F(A, B, C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C} + ABC$$

3 Variables (A, B, C) =  $2^3 = 8$  cells

|   |            |                  |            |            |    |                  |            |            |
|---|------------|------------------|------------|------------|----|------------------|------------|------------|
|   | BC         | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | BC | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ |
| A | $\bar{A}0$ | 0                | 1          | 2          | 3  | 4                | 5          | 6          |
| A | A1         | 1                | 5          | 1          | 7  | 1                | 6          | 1          |

Group 1:  $\bar{A}\bar{B}C$   
 Group 2:  $\bar{A}B\bar{C}$   
 Group 3:  $AB\bar{C}$

Group 1 (A)  $\bar{B} \cdot \bar{C} = \bar{A}\bar{B}C$   
 (BC)  $\bar{B}C$

Group 2 (A)  $\bar{A} \cdot \bar{C} = \bar{A}B\bar{C}$   
 (BC)  $B\bar{C}$

Group 3 (A)  $A$   
 (BC)  $B\bar{C} \cdot B\bar{C} = AB\bar{C}$

$$F(A, B, C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C}$$

$$3) Y = \sum m(0, 1, 3, 7)$$

Value = (0-7) 3 Variable = 8 Cells.

|   |           |                  |            |            |    |
|---|-----------|------------------|------------|------------|----|
|   | BC        | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | BC |
| A | $\bar{A}$ | 1                | 1          | 1          | 2  |
| A | A         | 1                | 5          | 1          | 6  |

Group 1:  $\bar{A}\bar{B}C$   
 Group 2:  $\bar{A}B\bar{C}$

Group 1 (A)  $\bar{A} \cdot A = 1$   
 (BC)  $BC = BC$

Group 2 (A)  $\bar{A}$   
 (BC)  $\bar{B}\bar{C} \cdot \bar{B}\bar{C} = \bar{A}\bar{B}$

$$Y = \bar{A}\bar{B} + BC$$

$$4) Y = m_3 + m_4 + m_6 + m_7$$

Value (0-7) 3 Variable = 8 cells

| A \ BC    | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | $BC$ |
|-----------|------------------|------------|------------|------|
| $\bar{A}$ | 0                | 1          | 3          | 2    |
| A         | 4                | 5          | 7          | 6    |

Group 1 (A)  $\bar{A} \cdot A = 1$   
 $BC \quad BC \quad = 1 \cdot BC = BC$

Group 2 (A)  $A$   
 (BC)  $\bar{B}\bar{C} \cdot \bar{B}\bar{C} = A\bar{C}$

$$Y(A, B, C) = BC + A\bar{C}$$

5)  $F(A, B, C) = \bar{A}\bar{B} + \bar{A}B + \bar{A}BC + BC$  Convert the expression to standard SOP form.

$$\begin{aligned} F(A, B, C) &= \bar{A}C(B + \bar{B}) + \bar{A}B(C + \bar{C}) + \bar{A}BC + BC(A + \bar{A}) \\ &= \bar{A}BC + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}B\bar{C} + \bar{A}BC + ABC + \bar{A}BC \\ &= \bar{A}BC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC \end{aligned}$$

| A \ BC    | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | $BC$ |
|-----------|------------------|------------|------------|------|
| $\bar{A}$ | 0                | 1          | 3          | 2    |
| A         | 4                | 5          | 7          | 6    |

1) Simplify the POS expression using K-map method

$$F = \Pi M(0, 6, 7, 8, 12, 13, 14, 15)$$

| AC \ CD | 00 | 01 | 10 | 11 |
|---------|----|----|----|----|
| AB      | 0  | 1  | 1  | 1  |
| AB      | 1  | 1  | 0  | 0  |
| AB      | 0  | 0  | 0  | 0  |
| AB      | 0  | 1  | 1  | 1  |

Group 1:  $\overline{A}\overline{B}$   $\overline{A}\overline{B}$   $\overline{A}\overline{B}$   $\overline{B}$

Group 2:  $\overline{C}\overline{D}$   $\overline{C}\overline{D}$   $\overline{C}\overline{D}$   $\overline{C}\overline{D}$

Group 3:  $\overline{A}B$   $\overline{A}B$   $\overline{A}B$   $BC$

$$F = \overline{C}\overline{D}\overline{B} + \overline{A}\overline{B} + BC$$

Applying de Morgan's theorem.

$$F = (C + D + B)(\overline{A} + \overline{B})(\overline{B} + \overline{C})$$

2) Simplify  $F(A, B, C) = \overline{A}BC + B\overline{C} + AB\overline{C} + A\overline{B}C$  in POS & SOP Forms.

Convert to standard SOP form

$$F(A, B, C) = \overline{A}BC + B\overline{C}(A + \overline{A}) + AB\overline{C} + A\overline{B}C$$

$$= \overline{A}BC + A\overline{B}C + \overline{A}B\overline{C} + AB\overline{C}$$

Group 1  $A \quad \bar{A}$   
 $BC \quad \bar{B}C \quad B\bar{C} \quad C$

Group 2  $\bar{A}$   
 $BC \quad B\bar{C} \quad \bar{A}B$

$$F(A, B, C) = \bar{A}B + C$$

6) Use a k-map to minimize the following SOP expression

$$F(A, B, C, D) = A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D} \\
+ \bar{A}\bar{B}CD + A\bar{B}CD + A\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} \\
+ \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D}$$

| AB \ CD          | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$ | $C\bar{D}$ |
|------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 1                |            | 1    | 1          |
| $\bar{A}B$       | 1                |            |      | 1          |
| $AB$             | 1                |            |      | 1          |
| $A\bar{B}$       | 1                |            | 1    | 1          |

Group 1  $\bar{A}\bar{B} \quad \bar{A}B \quad AB \quad A\bar{B}$   
 $\bar{C}\bar{D} \quad \bar{C}D \quad CD \quad C\bar{D} = \bar{D}$

Group 2  $\bar{A}\bar{B} \quad A\bar{B}$   
 $CD \quad C\bar{D} = \bar{B}C$

$$F(A, B, C, D) = \bar{D} + \bar{B}C$$

7) Simplify the expression

$$F = \sum m(0, 2, 4, 5, 6, 7, 8, 10, 11, 12, 14, 15)$$

(3 groups)

$$F = \bar{D} + \bar{A}B + AC$$

8) Simplify the expression using K-Map

$$F = m_1 + m_5 + m_{10} + m_{11} + m_{12} + m_{13} + m_{15}$$

$$F = AB\bar{C} + \bar{A}\bar{C}D + ACD + A\bar{B}C$$

9) Simplify the following SOP expression on a K-map.

$$Y = \bar{B}\bar{C} + A\bar{B} + AB\bar{C} + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}CD$$

$$F(A, B, C, D) = \bar{A}\bar{B}\bar{C} + A\bar{C} + AB$$

Simplification of POS Expression

1) Simplify the POS expression using K-map Method

$$F = \Pi M(0, 1, 4, 5, 6, 8, 9, 12, 13, 14)$$

| AB \ CD          | 00 | 01 | 10 | 11 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0  | 0  |    |    |
| $\bar{A}B$       | 0  | 0  |    | 0  |
| $A\bar{B}$       | 0  | 0  |    | 0  |
| $AB$             | 0  | 0  |    |    |

Group 1

|    |                  |            |            |      |
|----|------------------|------------|------------|------|
| AB | $\bar{A}\bar{B}$ | $\bar{A}B$ | $A\bar{B}$ | $AB$ |
| CD | $\bar{C}\bar{D}$ | $\bar{C}D$ | $C\bar{D}$ | $CD$ |

Group 2

|    |                  |            |
|----|------------------|------------|
| AB | $\bar{A}\bar{B}$ | $\bar{A}B$ |
| CD | $\bar{C}\bar{D}$ | $C\bar{D}$ |

$$F = \bar{C} + B\bar{D} \text{ (SOP)}$$

$$F = \bar{C} + (B + \bar{D})$$

$$F = C \cdot (\bar{B} + D)$$

2) Simplify the POS expression using K-map method.

$$F = \Pi M(0, 6, 7, 8, 12, 13, 14, 15)$$

| AB \ CD          | $\bar{C}\bar{D}$ | $\bar{C}D$ | $C\bar{D}$ | $CD$ |
|------------------|------------------|------------|------------|------|
| $\bar{A}\bar{B}$ | 0                | 1          | 3          | 2    |
| $\bar{A}B$       | 4                | 5          | 7          | 6    |
| $AB$             | 12               | 13         | 15         | 14   |
| $A\bar{B}$       | 8                | 9          | 11         | 10   |

Group 1

| AB | $\bar{A}\bar{B}$ | $A\bar{B}$ | $\bar{B}$ |
|----|------------------|------------|-----------|
| CD | $\bar{C}\bar{D}$ |            |           |

Group 2

| AB | $\bar{A}\bar{B}$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | $C\bar{D}$ | $CD$ |
|----|------------------|------------------|------------|------------|------|
| CD | $\bar{A}B$       |                  |            |            |      |

Group 3

| AB | $\bar{A}\bar{B}$ | $A\bar{B}$ |      |
|----|------------------|------------|------|
| CD | $\bar{C}\bar{D}$ | $\bar{C}D$ | $BC$ |

$$F = \bar{C}\bar{D}\bar{B} + AB + BC$$

applying de Morgan's theorem.

$$F = (C + D + B)(\bar{A} + \bar{B})(\bar{B} + \bar{C})$$

3) Simplify  $F(A, B, C) = \bar{A}BC + B\bar{C} + AB\bar{C} + A\bar{B}C$  in POS & SOP Forms.

Convert to standard SOP form

$$\begin{aligned} F(A, B, C) &= \bar{A}BC + B\bar{C}(A + \bar{A}) + AB\bar{C} + A\bar{B}C \\ &= \bar{A}BC + AB\bar{C} + \bar{A}B\bar{C} + A\bar{B}C \end{aligned}$$

| A \ BC    | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | $BC$ |
|-----------|------------------|------------|------------|------|
| $\bar{A}$ | 0                | 0          | 1          | 1    |
| A         | 0                | 1          | 0          | 1    |

POS form:

Group 1       $A$        $\bar{A}$   $A$   
                   $BC$        $\bar{B}\bar{C}$

Group 2       $\bar{A}\bar{B}C$

Group 3       $ABC$

$$F = \bar{B}\bar{C} + \bar{A}\bar{B}C + ABC$$

apply De Morgan's Law

$$F = (B+C)(A+B+\bar{C})(\bar{A}+\bar{B}+\bar{C})$$

SOP form:  $F = \bar{A}B + B\bar{C} + A\bar{B}C$

Simplify SOP expression using K-map

$$1) F(A, B, C, D) = \sum m(1, 4, 5, 6, 12, 14, 15)$$

$$F = B\bar{D} + \bar{A}\bar{C}D + ABC$$

$$2) F(A, B, C, D) = \sum (0, 1, 2, 4, 5, 7, 11, 15)$$

$$F = \bar{A}\bar{C} + \bar{A}B\bar{D} + ACD + \bar{A}\bar{B}\bar{D}$$

$$3) F(A, B, C, D) = \sum (2, 3, 10, 11, 12, 13, 14, 15)$$

$$F = AB + \bar{B}C$$

$$4) F(A, B, C, D) = \sum (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

$$F = \bar{A}B + BD + \bar{B}\bar{D}$$

5.  $F(x, y, z) = \sum m(1, 2, 3, 4, 5, 7)$  with NAND gates

$$F = z + \bar{x}y + x\bar{y}$$

6.  $F(A, B, C) = \sum m(0, 1, 3, 7) + \sum d(2, 5)$

$$F = C + \bar{A}\bar{B}$$

7.  $F(w, x, y, z) = \sum m(0, 7, 8, 9, 10, 12) + \sum d(2, 5, 13)$

$$F = w\bar{x} + \bar{x}\bar{z} + \bar{w}x$$

Don't Care Condition

1) In some condition digital system, certain output conditions never occur during normal operations is called "Don't Care conditions".

2) Therefore the output never appears is indicated by an "x" in the truth table.

1) Simplify using K-map

$$F(A, B, C, D) = \sum m(7, 8, 9) + \sum d(10, 11, 12, 13, 14, 15)$$

| AB \ CD          | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$ | $C\bar{D}$ |
|------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0                | 1          | 2    | 3          |
| $\bar{A}B$       | 4                | 5          | 6    | 7          |
| $AB$             | 8                | 9          | 10   | 11         |
| $A\bar{B}$       | 12               | 13         | 14   | 15         |

$$F(A, B, C, D) = A + BCD$$

2) Simplify using K-map  $F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + \sum d(0, 2, 5)$

| AB \ CD | 00             | 01             | 11              | 10              |
|---------|----------------|----------------|-----------------|-----------------|
| 00      | X <sub>0</sub> | 1 <sub>1</sub> | 1 <sub>3</sub>  | X <sub>2</sub>  |
| 01      |                | X <sub>4</sub> | 1 <sub>7</sub>  |                 |
| 11      |                |                | 1 <sub>5</sub>  |                 |
| 10      |                |                | 1 <sub>11</sub> | 1 <sub>15</sub> |

$$F = \bar{A}\bar{B} + CD$$

3) Using the K-map method, simplify the following boolean function  
 $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7) + \sum d(8, 10, 11, 15)$

| AB \ CD | 00             | 01 | 11              | 10              |
|---------|----------------|----|-----------------|-----------------|
| 00      | 1 <sub>0</sub> |    | 1 <sub>3</sub>  | 1 <sub>2</sub>  |
| 01      |                |    | 1 <sub>7</sub>  | 1 <sub>6</sub>  |
| 11      |                |    | X <sub>15</sub> |                 |
| 10      | X <sub>8</sub> |    | X <sub>11</sub> | X <sub>10</sub> |

$$F = C\bar{A} + \bar{D}\bar{B}$$

4) Simplify the expression

$$F(A, B, C, D) = \sum m(4, 5, 6, 7, 8, 12) + \sum d(1, 2, 3, 9, 11, 14)$$

|                  |    | CD               |                |                 |                 |
|------------------|----|------------------|----------------|-----------------|-----------------|
| AB               | CD | $\bar{C}\bar{D}$ | $\bar{C}D$     | $CD$            | $C\bar{D}$      |
|                  |    |                  |                |                 |                 |
| $\bar{A}\bar{B}$ |    | 0                | X <sub>1</sub> | X <sub>3</sub>  | X <sub>2</sub>  |
| $\bar{A}B$       |    | 0 <sub>4</sub>   | 0 <sub>5</sub> | 0 <sub>7</sub>  | 0 <sub>6</sub>  |
| $AB$             |    | 0 <sub>12</sub>  |                |                 | X <sub>14</sub> |
| $A\bar{B}$       |    | 0 <sub>8</sub>   | X <sub>9</sub> | X <sub>11</sub> |                 |

Group 1  $\bar{A}B$   $\bar{A}\bar{B}$

Group 2  $AB$   $A\bar{B}$   $A\bar{B} = A$   
 $CD$   $\bar{C}\bar{D}$

$$F(A, B, C, D) = \bar{A}B + A\bar{C}\bar{D}$$

applying de Morgan's Theorem.

$$F(A, B, C, D) = (A + \bar{B})(\bar{A} + \bar{C} + \bar{D})$$

5) Simplify the POS expression

$$F(A, B, C, D) = \Pi M(0, 3, 4, 7, 8, 10, 12, 14)$$

$$d(2, 6)$$

| AB               | CD | $\bar{C}\bar{D}$ | $\bar{C}D$ | $CD$           | $C\bar{D}$      |
|------------------|----|------------------|------------|----------------|-----------------|
|                  |    |                  |            |                |                 |
| $\bar{A}\bar{B}$ |    | 0 <sub>4</sub>   |            | 0 <sub>3</sub> | X <sub>2</sub>  |
| $\bar{A}B$       |    | 0 <sub>1</sub>   |            | 0 <sub>7</sub> | X <sub>6</sub>  |
| $AB$             |    | 0 <sub>12</sub>  |            |                | 0 <sub>14</sub> |
| $A\bar{B}$       |    | 0 <sub>8</sub>   |            |                | 0 <sub>10</sub> |

Group 1  $AB = 1$

$CD$   $\bar{C}\bar{D}$   $\bar{C}D$   $C\bar{D}$   $C\bar{D} = \bar{D}$

Group 2  $AB$   $\bar{A}\bar{B}$   $\bar{A}B = \bar{A}$

$CD$   $CD$   $C\bar{D} = C$

$$F = \bar{D} + AC$$

$$F(A, B, C, D) = D(A + \bar{C})$$

Quine Mc-cluskey Method of Minimization  
of logic functions or Tabulation Method.

i) Simplify the following boolean for  
 by using the Quine McCluskey Method.

$$F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$$

Step 1: List all minterms in the binary  
 form.

| Minterms | Binary representation |
|----------|-----------------------|
| $m_0$    | 0000                  |
| $m_2$    | 0010                  |
| $m_3$    | 0011                  |
| $m_6$    | 0110                  |
| $m_7$    | 0111                  |
| $m_8$    | 1000                  |
| $m_{10}$ | 1010                  |
| $m_{12}$ | 1100                  |
| $m_{13}$ | 1101                  |

Step 2: Arrange the minterms according to  
 the number of 1's

|         | Minterms | Binary representation |         |
|---------|----------|-----------------------|---------|
| Group 1 | $m_0$    | 0000                  | Group 1 |
|         | $m_2$    | 0010                  |         |
| Group 2 | $m_8$    | 1000                  | Group 2 |
|         | $m_3$    | 0011                  |         |
|         | $m_6$    | 0110                  |         |
| Group 3 | $m_{10}$ | 1010                  | Group 3 |
|         | $m_{12}$ | 1100                  |         |
| Group 4 | $m_7$    | 0111                  |         |
|         | $m_{13}$ | 1101                  |         |

Step 3: Compare each binary number with every term in the next category. If they differ only by one position, use a "-" against the binary representation of minterms.

| Minterms | Binary representation |      | Comparison     |
|----------|-----------------------|------|----------------|
| 0, 2     | 0000                  | 0010 | 00 - 0         |
| 0, 8     | 0000                  | 1000 | - 000          |
| 2, 3     | 0010                  | 0011 | 001 -          |
| 2, 6     | 0010                  | 0110 | 0 - 10         |
| 2, 10    | 0010                  | 1010 | - 010          |
| 2, 12    | 0010                  | 1100 | Not comparable |
| 8, 3     | 1000                  | 0011 | Not comparable |
| 8, 6     | 1000                  | 0110 | Not comparable |
| 8, 10    | 1000                  | 1010 | 10 - 0         |
| 8, 12    | 1000                  | 1100 | 1 - 00         |
| 3, 7     | 0011                  | 0111 | 0 - 11         |
| 3, 13    | 0011                  | 1101 | Not comparable |

|        |      |      |                |
|--------|------|------|----------------|
|        |      |      |                |
| 6, 7   | 0110 | 0111 | 011 -          |
| 6, 13  | 0110 | 1101 | Not comparable |
| 10, 7  | 1010 | 0111 | Not comparable |
| 10, 13 | 1010 | 1101 | Not comparable |
| 12, 7  | 1100 | 0111 | Not comparable |
| 12, 13 | 1100 | 1101 | 110 -          |

Step 4: Apply the same process described in step 3 until no further elimination of literals:

| Minterms | Binary represent | Minterm Pairs | Comparison |
|----------|------------------|---------------|------------|
| 0, 2     | 00 - 0           | 0, 2, 8, 10   | - 0 - 0    |
| 0, 8     | - 000            | 0, 8, 2, 10   | - 0 - 0    |
| 2, 3     | 001 -            | 2, 3, 6, 7    | 0 - 1 -    |
| 2, 6     | 0 - 10           | 2, 6, 3, 7    | 0 - 1 -    |
| 2, 10    | - 010            |               |            |
| 8, 10    | 10 - 0           |               |            |
| 8, 12    | 1 - 00           |               |            |
| 3, 7     | 0 - 11           | 8, 12, 7      | 1 - 00     |
| 6, 7     | 0 11 -           | 12, 13        | 1 1 0 -    |
| 12, 13   | 110 -            | missing       |            |

Step 5: List the primary implicants

| Primary Implicants | Binary representation |
|--------------------|-----------------------|
| 0, 2, 8, 10        | - 0 - 0               |
| 2, 3, 6, 7         | 0 - 1 -               |
| 8, 12              | 1 - 0 0               |
| 12, 13             | 1 1 0 -               |

step 6: Select the prime implicants which must cover all the minterms.

| Minterms<br>column<br>Prime<br>Implicants<br>No's | m <sub>0</sub> | m <sub>2</sub> | m <sub>3</sub> | m <sub>6</sub> | m <sub>7</sub> | m <sub>8</sub> | m <sub>10</sub> | m <sub>12</sub> | m <sub>13</sub> |
|---------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|
| ✓ 0, 2, 8, 10                                     | •              | •              |                |                |                | •              | •               |                 |                 |
| ✓ 2, 3, 6, 7                                      |                | •              | •              | •              | •              |                |                 |                 |                 |
| 8, 12                                             |                |                |                |                |                | •              |                 | •               |                 |
| ✓ 12, 13                                          |                |                |                |                |                |                |                 | •               | •               |

Step 7: List the selected prime implicants and its corresponding expression.

| Selected Minterms | Binary representation                                        | Expression        |
|-------------------|--------------------------------------------------------------|-------------------|
| (0, 2, 8, 10)     | $\begin{matrix} A & B & C & D \\ - & 0 & - & 0 \end{matrix}$ | $\bar{B} \bar{D}$ |
| (2, 3, 6, 7)      | $0 - 1 -$                                                    | $\bar{A} C$       |
| (12, 13)          | $1 1 0 -$                                                    | $A B \bar{C}$     |

$$F(A, B, C, D) = \bar{B} \bar{D} + \bar{A} C + A B \bar{C}$$

2) Simplify the boolean function using Quine - McCluskey (tabulation method)

$$F = \sum (0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

3) Find the minimal sum of products for the boolean expression.

$$F(w, x, y, z) = \sum (1, 3, 4, 5, 9, 10, 11) + \sum (6, 8)$$

using the Quine - McCluskey method

4) Simplify the following Boolean function by using tabulation method

$$F(w, x, y, z) = \sum (2, 3, 12, 13, 14, 15)$$

5) Simplify the boolean function using tabulation method  $F = \sum (0, 1, 4, 11, 15, 32, 29, 28)$

6) Implementation of boolean function using universal gates & Multilevel gate

### Universal Gates

Universal logic gates are the logic gates that are capable of implementing any boolean function without requiring any other type of gate.

They are called "Universal gates" because

1. They can realize all the binary operations

2. All the basic logic gates can be derived from them.

They have the following properties

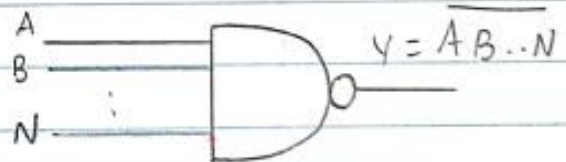
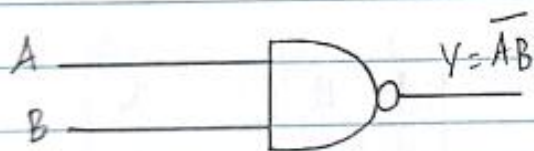
1. Universal gates are not associative in nature
2. Universal gates are commutative in nature

There are following two universal logic gates

1. NAND gate
2. NOR gate

NAND gate

A NAND gate is constructed by connecting a NOT gate at the output terminal of the AND gate.



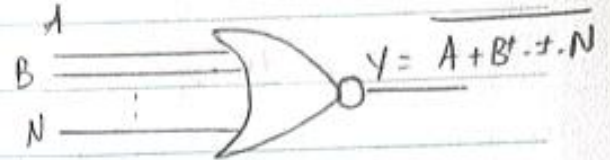
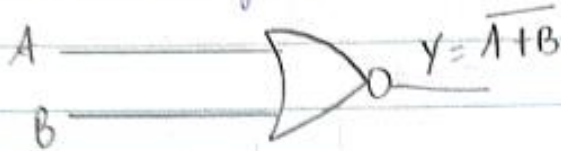
2 Input NAND gate

N-Input NAND gate

| A | B | $Y = (A \cdot B)'$ |
|---|---|--------------------|
| 0 | 0 | 1                  |
| 0 | 1 | 1                  |
| 1 | 0 | 1                  |
| 1 | 1 | 0                  |

## NOR Gate

A NOR gate is constructed by connecting a NOT gate at the output terminal of the OR gate.

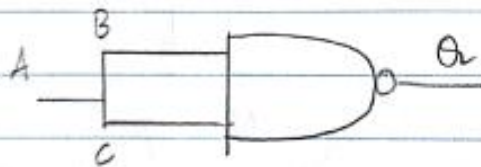


2 Input NOR gate

N - Input NOR gate

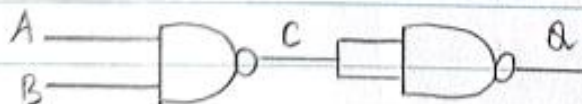
| A | B | $Y = (A + B)'$ |
|---|---|----------------|
| 0 | 0 | 1              |
| 0 | 1 | 0              |
| 1 | 0 | 0              |
| 1 | 1 | 0              |

1) How to use NAND gates to build a NOT gate?



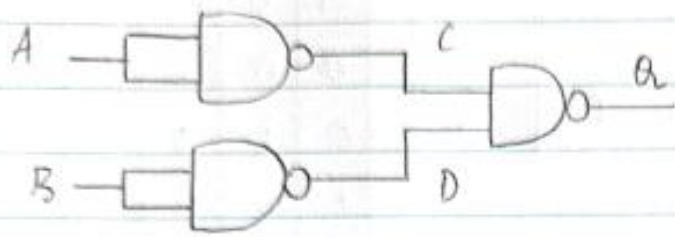
| A | B | C | Q |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

2) How to use NAND gate to build a AND gate?



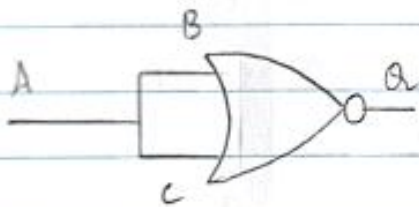
| A | B | C | Q |
|---|---|---|---|
| 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

3) How to use NAND gates to build an OR gate?



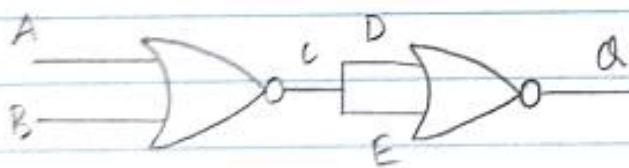
| A | B | C | D | A |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |

4) How to use NOR gate to build a NOT gate?



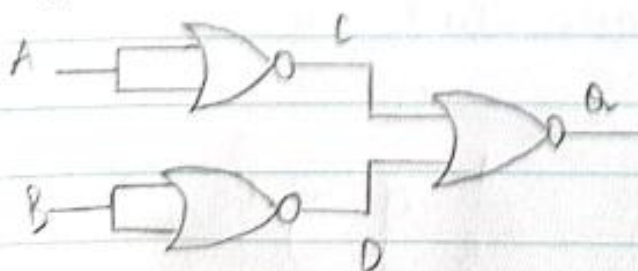
| A | B | C | A |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

5) How to use NOR gate to build an OR gate?



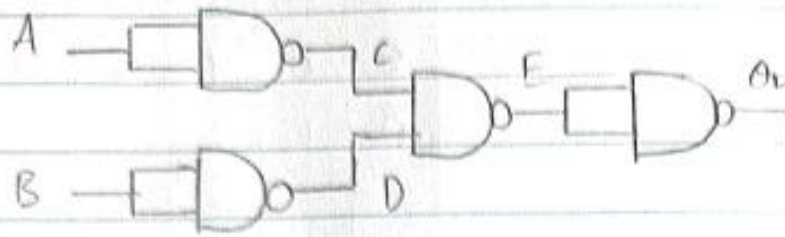
| A | B | C | D | E | A |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 |

6) How to use NOR gate to build an AND gate?



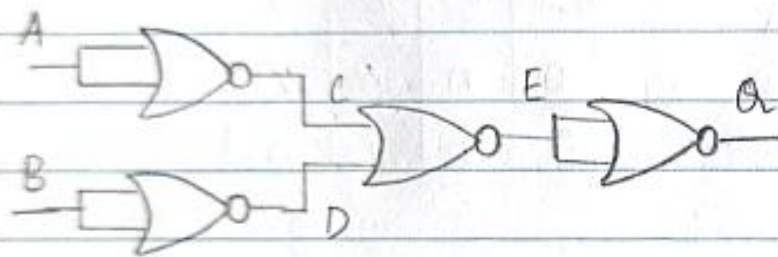
| A | B | C | D | A |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 |

7) How to use NAND gates to build a NOR gate?



| A | B | C | D | E | A |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 |

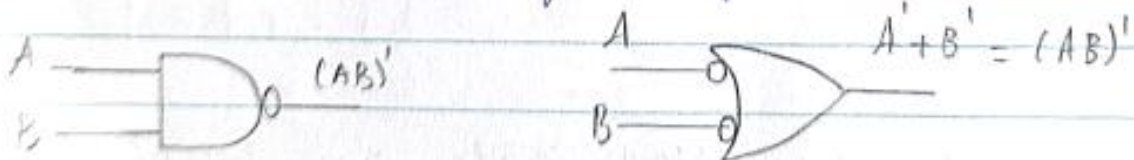
8) How to use NOR gate to build an NAND gate



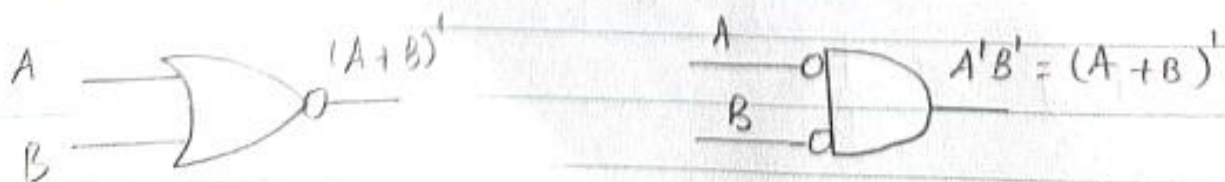
| A | B | C | D | E | A |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 | 0 |

## Equivalent Gates

1. A NAND gate is equivalent to an inverted-input OR gate



2. A NOR gate is equivalent to an inverted-input AND gate



## Two-level Implementations

1. Boolean function in either SOP or POS forms can be implemented using 2-Level implementations.

2. For SOP form AND gates will be in the first level and a single OR gate will be in the second level.

3. For POS form OR gates will be in the first level and a single AND gate will be in second level.

4. Using inverters to complement input variables is not counted as a level.

5. SOP forms can be implemented using only NAND gates, while POS forms can be implemented using NOR gates.

## Applications

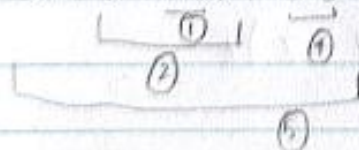
Used in manufacturing logic circuits devices such as multiplexers, registers, ALUs, Flash memory.

Multilevel gate:

$$2) F = AB + C + BC$$

$$F = (A+B)(C+D)$$

$$F = A(B+C+D) + \bar{C}D$$



## Important Question

2 Marks :-

1. Implement AND and OR gates using NAND gate
2. Perform  $(147 - 89)$  using 2's complement binary arithmetic
3. Subtract 321 from 550 using 10's and 9's complement.
4. Convert  $(0.513)_{10}$  to an Octal number
5. Why are NAND and NOR gate known as universal gates.
6. State De Morgan's theorem
7. Convert  $F(x, y, z) = \Sigma(1, 3, 7)$  to the other canonical form.
8. Define excess-3 code.
9. Convert Gray number 1110 to its BCD equivalent
10. Convert the following number from one base to other.
  - a)  $(354.52)_6 = (?)_{10}$
  - b)  $(65.342)_8 = (?)_7$
11. What are error detecting codes.
12. Find the complement for the following function.
 
$$F1 = xy' + x'y'$$

$$F2 = (xy + y'z + xz)'$$

11 Marks:

1) Simplify using Karnaugh map

$$F(A, B, C, D) = \sum m(7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$

2) Convert the following number as stated.

a)  $4658_{10}$  to binary

b)  $110110110111_2$  to Hexadecimal

c)  $10010111_2$  to decimal

d)  $AEFB_{16}$  to Octal

e)  $0.263_{10}$  to binary.

3) Simplify the boolean function using Quine - McCluskey (tabulation) Method

$$F = \sum(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

4) Explain the excess-3 to BCD Converter

5) Convert the following numbers to the indicated bases. <sup>decimal 25</sup> Binary 11010111 to binary, octal + hexa  
Binary 11010111 to decimal, octal, hexa

6) Explain the algorithm for subtraction using 10's complement method and also find out the following using the same

$$a) (72532 - 3250)_{10}$$

$$b) (3250 - 72532)_{10}$$

7) Simplify the boolean function F using Karnaugh map method

$$F(A, B, C, D) = \sum(1, 4, 5, 6, 12, 14, 15)$$

$$F(A, B, C, D) = \sum(0, 1, 2, 4, 5, 7, 11, 16)$$

$$F(A, B, C, D) = \sum(2, 3, 10, 11, 12, 13, 14, 16)$$

$$F(A, B, C, D) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

8) Find the minimal sum of products for the boolean expression.

$$F(W, X, Y, Z) = \sum(1, 3, 4, 5, 9, 10, 11) + \sum(6, 8) \text{ using the Quine - McCluskey Method and also draw the logic circuit diagram.}$$